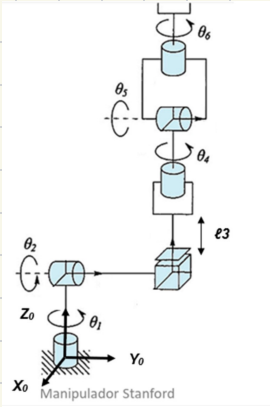

$A_1 \to A_2$

$\boxed{\text{Rotacion}}$ de 90° positiva en X

$$\begin{bmatrix} \cos(\theta_1) & -\text{sen}(\theta_1) & 0 \\ \text{sen}(\theta_1) & \cos(\theta_1) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \cos(\theta_1) & 0 & \sin(\theta_1) \\ \sin(\theta_1) & 0 & -\cos(\theta_1) \\ 0 & 1 & 0 \end{bmatrix}$$

$\boxed{\text{Traslacion}}$

$$\begin{bmatrix} 0 \\ 0 \\ l_1 \end{bmatrix}$$

$A_2 \to A_3$

$\boxed{\text{Rotacion}}$ de 90° Negativa en X

$$\begin{bmatrix} \cos(\theta_4) & -\text{sen}(\theta_4) & 0 \\ \text{sen}(\theta_4) & \cos(\theta_4) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -\sin(\theta_4) & -\cos(\theta_4) \\ 0 & \cos(\theta_4) & -\sin(\theta_4) \\ 1 & 0 & 0 \end{bmatrix}$$

$\boxed{\text{Traslacion}}$

$$\begin{bmatrix} 0 \\ 0 \\ -h \end{bmatrix}$$

$A_3 \to A_4$

$\boxed{\text{Rotacion}}$ no hay

$\boxed{\text{Traslacion}}$

$$\begin{bmatrix} 0 \\ 0 \\ l_3 + t \end{bmatrix}$$

$A_4 \to A_5$

$\boxed{\text{Rotacion}}$ de 90° positiva en X

$$\begin{bmatrix} \cos(\theta_1) & -\text{sen}(\theta_1) & 0 \\ \text{sen}(\theta_1) & \cos(\theta_1) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \cos(\theta_1) & 0 & \sin(\theta_1) \\ \sin(\theta_1) & 0 & -\cos(\theta_1) \\ 0 & 1 & 0 \end{bmatrix}$$

$\boxed{\text{Traslacion}}$

$$\begin{bmatrix} 0 \\ 0 \\ l_4 \end{bmatrix}$$